

Experimental Evaluation of PAM2-SSB and PAM2-SSB+DCPC for a 2×100Gb/s/λ Very High Speed PON Downstream

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Abstract We experimentally demonstrate a digitally-generated PAM2-SSB modulation with direct-detection receiver. The results show that, even with limited DSP complexity, the CD tolerance can be enhanced to ensure coexistence with previous PON technologies while still supporting Class N1 power-budget. Further improvement is obtained by adding DCPC. ©2026 The Author(s)

Introduction

The latest recommendation for passive optical networks (PON) published by the ITU-T is 50G-PON, which is now entering the early stages of commercial deployment [1]. Thus, an exploratory assessment of potential technical candidates for the next PON evolution, referred as very high speed PON (VHSP), has started [1]. VHSP is expected to ensure operation over deployed optical distribution networks (ODNs) while supporting the coexistence of multiple PON generations and to deliver a service capacity of at least 200 Gb/s. This capacity requirement implies a line rate significantly higher than 200 Gb/s to accommodate the necessary protocol overheads [2].

While several experiments and simulations have demonstrated the feasibility of reusing intensity modulation with direct-detection (IM-DD) for VHSP, these are generally limited to operation near the nominal zero-dispersion wavelength of single-mode fibre (SMF) [3,4]. To simultaneously meet the requirements for line rate and multiple coexistence, digital chromatic dispersion pre-compensation (DCPC) has been proposed for downstream (DS) transmission [5]. This technique pre-equalizes the DS signal for a specific amount of CD using a complex-valued FIR-filter at the transmitter (TX), thus requiring an optical IQ modulator (IQM). Although this introduces complexity, it is centralized at the optical line terminal (OLT), thereby maintaining a cost-effective DD receiver (RX) at the optical network unit (ONU). Since DCPC compensates for a fixed CD value, while ONUs distributed across the ODN experience variable accumulated CD, grouping ONUs based on their location or accumulated CD has been proposed, along with the use of a shared low-bitrate header for control and management [6].

In this paper, we investigate a digitally generated single sideband (SSB) PAM2 optical system to enhance the CD tolerance. While it was previously studied at 50 Gb/s [7], we experimentally scale this approach to 100 Gb/s and utilize a second wavelength to achieve the 200 Gb/s capacity

required for VHSP. Using DD and a simple equalizer at the ONU RX, the tolerable accumulated CD reaches up to 200 ps/nm for a 3 dB penalty. With stronger equalization, the CD tolerance increases to 250 ps/nm, and further extends to 350 ps/nm when combining DCPC with SSB, all while maintaining a class N1 power budget.

Experimental setup

A schematic of the experimental setup is depicted in Fig. 1. We generated a non-return to zero (NRZ) PRBS-11 sequence and applied a 15-tap asymmetric FIR filter to produce the Hilbert transform, with coefficients calculated according to [8]. When DCPC was enabled, a pair of 26-tap FIR filters, addressing the real and imaginary part respectively, were added with coefficients derived from [5]. These two filters could be effectively combined into a single complex-valued FIR. The resulting filtered real and imaginary signals were uploaded to a 128 GSa/s arbitrary waveform generator (AWG) with a bandwidth (BW) of 65 GHz. A second set of samples, generated with an identical filtering process but a different PRBS seed was uploaded to two separate channels. The outputs of two AWG channels were amplified by 70 GHz RF drivers and used to modulate the light of an external cavity laser (ECL) emitting at 1550 nm through an optical IQM with a 3 dB BW of 40 GHz. Due to the BW limitation of the IQM, the overall bitrate was restricted to 100 Gb/s/λ. We optimized the RF driving signals to obtain a carrier to signal power ratio (CSPR) of ~9 dB. The two remaining AWG channels modulated a second IQM, optically fed by a 1546 nm distributed feedback laser (DFB) in order to get a 2×PAM2-SSB modulated signal. Afterwards, the power levels of the two optical signals were balanced and combined using a 3 dB optical coupler. An erbium doped fibre amplifier (EDFA) boosted the combined signal, setting the launched power for each wavelength to +8 dBm (the signal spectra are shown as an inset in Fig. 1). The optical signal was then transmitted over various lengths of SMF with a measured average CD coefficient of

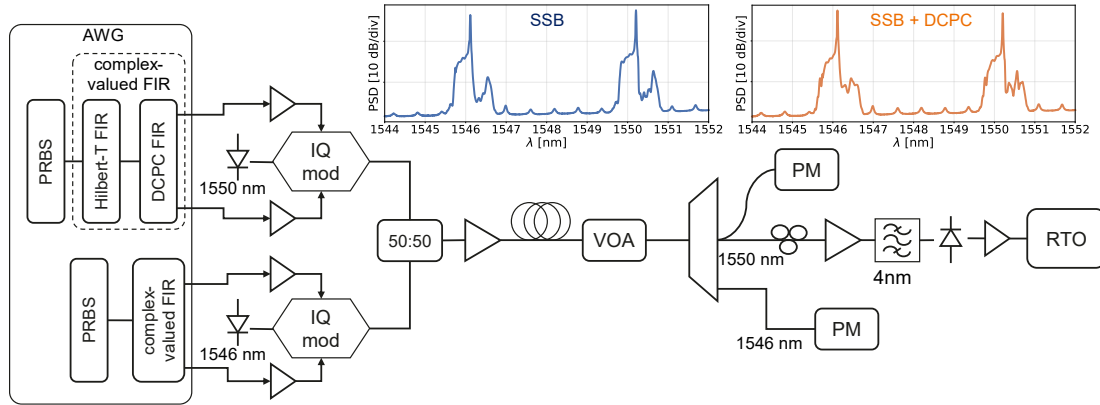


Fig. 1: Experimental setup for the 200 Gb/s DS. The insets show the optical spectrum for (left) SSB signals and (right) SSB+DCPC signals (PM: power meter)

approximately 17.4 ps/nm/km at 1550 nm. After transmission, a variable optical attenuator (VOA) emulated ODN losses and controlled the received power. This was followed by a CWDM demultiplexer with 4nm channel spacing used to separate the signals. The Rx consisted of an EDFA providing a gain of 20 dB and having noise figure (NF) of ~5 dB followed by a 4 nm optical filter (with IL $\approx$ 5 dB) centred at 1550 nm. When measuring the second optical channel, the filter was tuned to 1546 nm. The tuneable filter and the CWDM filter could be merged having a single SOA amplifying both channels. A 95 GHz p-i-n photodiode (PD) with responsivity of 0.5 A/W converted the optical signal to the electrical domain, followed by a 75 GHz RF amplifier. We note that this optically pre-amplified Rx could be integrated in an SOA-filter-PIN structure. The electrical signal was captured by a real-time oscilloscope (RTO) with a 110 GHz BW and sample rate of 256 GSa/s. The waveforms were processed offline, performing clock recovery, down-sampling to 1 Sa/symbol and equalisation with either an 11-tap or 23-tap FFE. The equalized samples were fed into a noise whitening filter followed by a 1-bit memory maximum likelihood sequence estimator (MLSE), after which the BER was calculated.

Results

We first tested a conventional NRZ signal (modulating only one MZM of the IQM), computing the Rx sensitivity after various fibre lengths at a pre-FEC BER threshold of $2 \cdot 10^{-2}$. Fig. 2 shows the optical path penalty (OPP) as a function of the accumulated CD. At a 3 dB penalty limit, we observe that the CD tolerance is restricted to ~40 ps/nm. Afterwards, we optimized the amplitude of the four AWG waveforms as well as the IQM bias points to generate an SSB optical signal. As seen in the left inset of Fig. 1, a residual spectral component remains on the suppressed sideband, primarily due to the limited sampling rate of the electrical signal.

The SSB penalty, referenced to the NRZ BtB

Rx sensitivity (-23.2 dBm), is shown in Fig. 2. Improved RX sensitivity is expected with an optimized RX [4]. For this analysis, we measured the BER at both optical wavelengths and plotted the average value. While SSB incurs a ~1.2 dB BtB penalty compared to NRZ, its CD tolerance is significantly enhanced, reaching up to 210 ps/nm at a 3 dB level. When scaling to 120 Gb/s bitrate (including 20% overhead), we account for the increased signal BW, noting that the CD-induced impairments approximately rise with the square of the BW expansion. Consequently, the accumulated CD tolerance reduces to ~145 ps/nm at a 3 dB penalty. Reserving a 1 dB margin for other impairments reduces the effective CD tolerance to 170 ps/nm at 100 Gb/s, corresponding to 118 ps/nm at 120 Gb/s at 1550 nm. Assuming O-band operation, we could transmit at a wavelength of 1370 nm, where the accumulated CD after a 20 km G.652 SMF link under worst-case CD is ~120 ps/nm. A key advantage over DCPC is that ONU grouping is unnecessary, as the entire CD range would be covered while enabling coexistence with existing PON technologies. We expect that with higher BW components than those used in this experiment, the Rx equalizer could be further simplified.

At a 3 dB penalty, we measured an Rx sensitivity of approximately -20.2 dBm. Although we anticipate this value could be improved by further optimizing the optical Rx frontend, the initial Tx

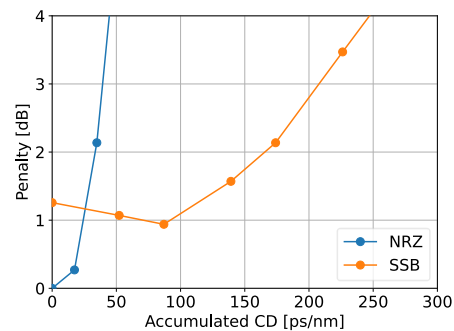


Fig. 2: Rx sensitivity penalty versus accumulated CD for 100 Gb/s NRZ and SSB signals.

launch power of +8 dBm per channel restricts the power budget to ~28 dB. This falls short of the lowest N1 class requirement (29 dB), particularly when accounting for the 0.5 dB IL of the CWDM filter. We thus increased the launch power to +10 dBm and +12 dBm per channel (total output power limited to +15 dBm) and evaluated the BER after 5 km transmission. The results, shown

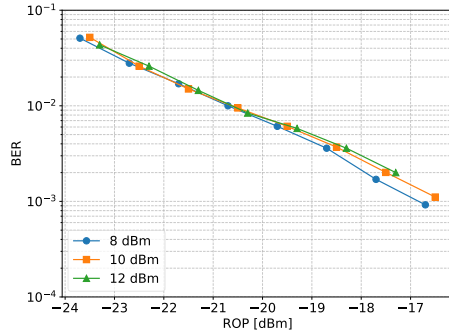


Fig. 3: BER versus ROP for 100Gb/s/λ SSB signal after 10 km transmission with different launch powers.

in Fig. 3, indicate overlapping curves with no penalty at a BER of $2 \cdot 10^{-2}$. However, we expect that optical launch powers exceeding +12 dBm per wavelength will induce nonlinearities such as self-phase modulation (SPM). Therefore, we constrained the launch power to +10 dBm, yielding a power budget of 29.5 dB.

To improve the system's CD tolerance, we increased the Rx FFE taps to 23 followed by MLSE while maintaining the Tx launch power at +10 dBm per channel. Further increasing the number of taps yielded negligible improvement. Fig. 4 illustrates the penalty relative to the NRZ BtB Rx sensitivity at a BER of $2 \cdot 10^{-2}$ (-23.3 dBm). For consistent comparison, the captured NRZ data was also processed with the stronger filter; however, the improvement for this modulation format was negligible. At a 3 dB penalty, the accumulated CD tolerance for the SSB signal increased to ~260 ps/nm, with a penalty remaining below 1 dB up to 180 ps/nm CD. Theoretically, scaling the latter value to a bitrate of 120 Gb/s results in an accumulated CD of ~125 ps/nm. Hence, an operating wavelength of 1370 nm could reach 20 km for the worst-case CD and achieve nearly class C+ power budget with a launch power of +10 dBm. For the same operating wavelength at 3 dB penalty, the fibre reach could be extended to 30 km while still meeting class N1 requirements.

The CD tolerance can be further improved by adding a DCPC FIR equalizer after the SSB FIR, merging them into a single complex-valued FIR filter to pre-compensate for a CD of 85 ps/nm. As with the SSB case, the Rx utilized a 23-tap FFE followed by 1-bit memory MLSE. The results,

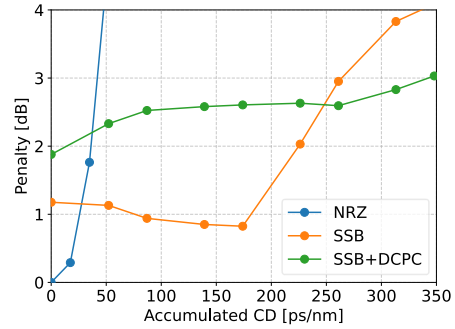


Fig. 4: Rx sensitivity penalty versus accumulated CD for 100Gb/s NRZ and SSB and SSB+DCPC signals, with 23-tap FFE followed by MLSE.

plotted against the NRZ BtB reference in Fig. 4 (green line), show that, while an additional BtB penalty of ~0.7 dB is incurred compared to SSB, the CD tolerance extends to 350 ps/nm at a 3 dB penalty. This 90 ps/nm improvement aligns closely with the pre-compensated value. Notably, the SSB+DCPC penalty curve remains nearly flat across the 80-120 ps/nm accumulated CD range. We attribute the higher penalty relative to SSB to the increased peak to average power ratio (PAPR) of the pre-compensated signal, which could be reduced through further optimization of the electrical drive signals. This accumulated CD tolerance translates to a potential operating wavelength of 1540 nm for a 20 km worst-case CD at 100 Gb/s. While this suggests feasibility in the C-band, the system would leave no room to accommodate other transmission penalties. However, by operating in the upper O-band, SSB modulation provides the required margin, still achieving the necessary power budget for PON.

Conclusions

We experimentally evaluated a PAM2-SSB modulation and a PAM2-SSB with DCPC for VHSP DS using two wavelengths operating at 100 Gb/s each. Utilizing an 11-tap FFE equalizer followed by MLSE, CD tolerance was significantly enhanced from ~40 ps/nm for conventional NRZ to over 200 ps/nm for SSB at a 3 dB penalty, despite ~1 dB BtB penalty associated with the latter. When scaled to 120 Gb/s/λ to achieve class N1 power budget, the highest possible operating wavelength (considering worst-case CD coefficient) would be approximately 1370 nm. Increasing the launch power to 10 dBm per wavelength showed no performance degradation, providing margin to accommodate other transmission impairments. In addition, PAM2-SSB CD tolerance was improved by using a 23-tap FFE and further enhanced with DCPC. Thus, PAM2-SSB can enable coexistence with previous PON technologies and offers the potential to further simplify the RX DSP.

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